

SESSION 5 • LECTURE 4

Thermal Equilibrium, Zeroth Law & the Concept of Temperature

Detailed lecture notes — equilibrium, the zeroth law, thermometric properties, solved numericals

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Learning objectives. After this session you should be able to: (i) define thermal equilibrium and the diathermic/adiabatic walls; (ii) state and interpret the zeroth law; (iii) explain how the zeroth law underpins temperature measurement; (iv) list common thermometric properties; (v) evaluate temperature from a linear thermometric property.

1. Thermal Equilibrium

When a hot body and a cold body are brought into communication through a heat-conducting (**diathermic**) wall, heat flows from hot to cold. The driving potential is the temperature difference; as it shrinks, the flow slows, and when it vanishes the bodies are in **thermal equilibrium** — no property of either body changes any further. An **adiabatic** wall (ideal insulation) prevents such interaction entirely.

Formally: *two systems are in thermal equilibrium if, when connected through a diathermic wall, no change in any observable property occurs.*

2. The Zeroth Law of Thermodynamics

If $A \equiv C$ (thermal equilibrium) and $B \equiv C$, then $A \equiv B$

R.H. Fowler, 1931 — logically prior to the First and Second Laws, hence “zeroth”

- Body C is the go-between: in practice, C is the **thermometer**.
- A and B need never touch each other; equality of their “thermal state” is established through C.
- The property shared by all bodies in mutual thermal equilibrium is defined as the **temperature**. Equal temperature $\Leftrightarrow$ thermal equilibrium $\Leftrightarrow$ no heat flow.
- Microscopic view: temperature measures the average kinetic energy of molecular motion.

KEY POINT The zeroth law is the logical foundation of all thermometry: it guarantees that a thermometer reading taken on one body can be meaningfully compared with a reading taken on another.

3. Thermometric Properties and Thermometers

Any property that changes measurably and monotonically with temperature can serve as a **thermometric property** x . The instrument built on it is calibrated at reproducible **fixed points** — classically the ice point and steam point of water; in the modern definition, the triple point of water (273.16 K).

Thermometer	Thermometric property x	Remarks
Liquid-in-glass (Hg, alcohol)	Length of liquid column, L	Simple, direct; -38 to 350 °C for mercury
Constant-volume gas thermometer	Pressure of fixed gas volume, p	Very wide range; the primary standard
Resistance thermometer (RTD)	Electrical resistance, R	Platinum; high accuracy, -200 to 850 °C
Thermocouple	Thermo-emf, ϵ	Fast, remote sensing, rugged
Constant-pressure gas thermometer	Volume of gas, V	Less common; same principle

The two-point linear scale

If x varies linearly with temperature, then with x_i at the ice point (0 °C) and x_s at the steam point (100 °C):

$$t = 100 \times (x - x_i) / (x_s - x_i) \text{ °C}$$

Works for any linear thermometric property: L , p , R or ϵ

4. Worked Examples

WORKED EXAMPLE 4.1 — Resistance thermometer

The resistance of a platinum wire is $10.00\ \Omega$ at the ice point and $13.86\ \Omega$ at the steam point. Resistance varies linearly with temperature. Find the temperature when $R = 12.40\ \Omega$.

Solution.

1. $t = 100 (R - R_i)/(R_s - R_i)$.
2. Numerator: $12.40 - 10.00 = 2.40\ \Omega$; denominator: $13.86 - 10.00 = 3.86\ \Omega$.
3. $t = 100 \times 2.40/3.86 = 62.2\ ^\circ\text{C}$.

ANSWER $t \approx 62.2\ ^\circ\text{C}$

WORKED EXAMPLE 4.2 — Thermocouple with linear emf

A thermocouple develops $0\ \text{mV}$ at $0\ ^\circ\text{C}$ and $4.10\ \text{mV}$ at $100\ ^\circ\text{C}$, varying linearly. What temperature corresponds to a reading of $2.87\ \text{mV}$, and what emf corresponds to $60\ ^\circ\text{C}$?

Solution.

1. $t = 100 \times (2.87 - 0)/(4.10 - 0) = 100 \times 0.70 = 70\ ^\circ\text{C}$.
2. Inverting: ε at $60\ ^\circ\text{C} = 4.10 \times 60/100 = 2.46\ \text{mV}$.

ANSWER $t = 70\ ^\circ\text{C}$ | $\varepsilon(60^\circ\text{C}) = 2.46\ \text{mV}$

WORKED EXAMPLE 4.3 — Gas thermometer at constant volume

A constant-volume gas thermometer reads a pressure of $65.0\ \text{kPa}$ at the triple point of water. What temperature corresponds to a pressure of $82.3\ \text{kPa}$?

Solution.

1. Ideal-gas scale: $T = 273.16 \times (p/p_{\text{tp}})$.
2. $T = 273.16 \times 82.3/65.0 = 273.16 \times 1.2662$.
3. $T = 345.9\ \text{K} \approx 72.7\ ^\circ\text{C}$.

ANSWER $T \approx 345.9\ \text{K}$ ($72.7\ ^\circ\text{C}$)

Note: This anticipates the absolute scale of Lecture 5.

5. Practice Problems (answers given)

#	Problem	Answer
P1	A mercury column is $4.0\ \text{cm}$ long at the ice point and $24.0\ \text{cm}$ at the steam point. Find t when $L = 13.0\ \text{cm}$.	$45\ ^\circ\text{C}$
P2	For the wire of Example 4.1, what resistance corresponds to $85\ ^\circ\text{C}$?	$13.28\ \Omega$
P3	Why does the zeroth law justify comparing readings of the same thermometer on two different bodies? (two lines)	Both equal C's state $\Rightarrow$ equal each other

6. Summary

- Thermal equilibrium: no heat flow; reached when the temperature difference vanishes.
- Zeroth law: $A \equiv C$ and $B \equiv C \Rightarrow A \equiv B$; body C is the thermometer.
- Temperature is the property whose equality defines thermal equilibrium.
- Linear thermometric property: $t = 100(x - x_i)/(x_s - x_i)\ ^\circ\text{C}$.

Text: P.K. Nag, "Engineering Thermodynamics", 6th Edn., Tata McGraw-Hill — Ch. 2 (Temperature). Reference: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.